

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I D.Anand Paul (192425207) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

D.Anand Paul

Signature of the student:

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

1. Concept Map: Security Attacks, Threats, Risks and Security Goals

Basic Concept Map



Security Attack	Threat Created	Risk	Security Goal	Real-World Example
Phishing	Stealing usernames/passwords	Account or data theft	Confidentiality	Fake banking email steals login details
Malware	Unauthorized access or data damage	Loss of sensitive information	Confidentiality + Integrity	Ransomware encrypts company files
Data modification	Unauthorized change of information	Incorrect or corrupted data	Integrity	Changing marks in a college database
DDoS attack	Making a service unavailable	Service interruption	Availability	Website becomes inaccessible due to huge traffic
Password attack	Unauthorized account access	Personal or business data exposure	Confidentiality	Attacker guesses a weak password

b. How security goals mitigate attacks

1 Confidentiality

Confidentiality ensures that information is accessible only to authorized users.

It can be protected using passwords, encryption, and access control.

Example: Banking applications encrypt customer information so that attackers cannot easily read stolen data.

2. Integrity

Integrity ensures that information is accurate and has not been changed without permission.

Hashing, digital signatures, and access controls help maintain integrity.

Example: A digital signature can verify that an important document has not been modified.

3. Availability

Availability ensures that systems and information are accessible whenever authorized users need them. Backups, redundancy, firewalls, and DDoS protection help maintain availability.

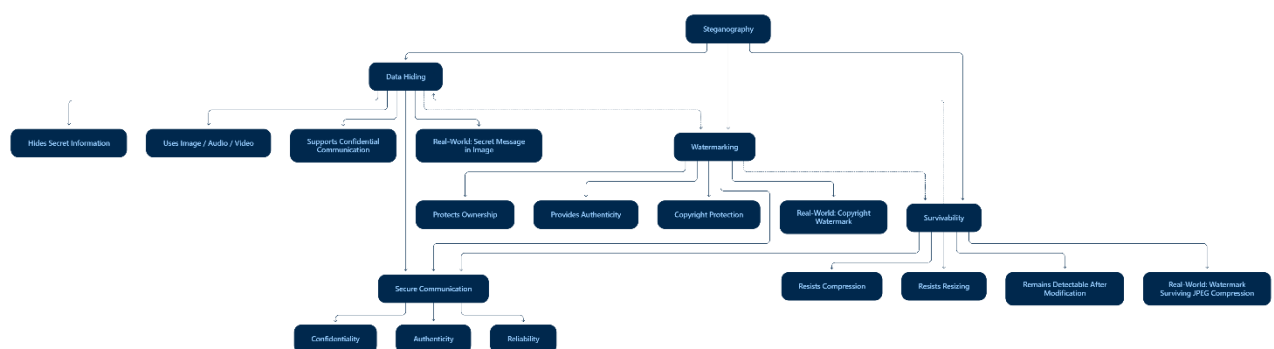
Example: A company uses backup servers so its website can continue working even if one server fails.

Conclusion

Security attacks can create threats that result in risks to an organization. The three major security goals—Confidentiality, Integrity, and Availability (CIA)—help reduce these risks by protecting information from unauthorized access, modification, and disruption.

2. Concept Map: Steganography – Data Hiding, Watermarking and Survivability

Concept Map



a. Relationship between the concepts

Data Hiding:

Data hiding is the main concept of steganography. It means placing secret information inside another normal-looking medium such as an image, audio file, or video.

Watermarking:

Watermarking is a special form of information embedding. It is mainly used to identify ownership, prove authenticity, or protect digital content.

Survivability:

Survivability refers to the ability of hidden information or a watermark to remain present and usable even after operations such as compression, resizing, or format conversion.

Therefore:

Data Hiding → Watermarking → Survivability

These concepts work together to make hidden communication more secure and reliable.

b. Contribution to secure communication

1. Data Hiding – Confidentiality

Data hiding makes the existence of a secret message difficult to notice.

Example: A confidential message can be hidden inside an ordinary image.

2. Watermarking – Authenticity and Ownership

Watermarking helps identify the owner of digital content and can help detect unauthorized use.

Example: A photographer adds a digital watermark to an image to show ownership.

3. Survivability – Reliability

Survivability ensures that the hidden information remains available even when the carrier file is modified.

Example: A watermark that survives image compression can still be detected after the image is uploaded to a website.

c. Real-world application for each

Concept	Purpose	Real-World Application
Data Hiding	Hide secret information	Confidential message hidden inside an image
Watermarking	Protect ownership/authenticity	Copyright watermark on digital photographs
Survivability	Preserve hidden data after modifications	Watermark remaining after JPEG compression of an image

Conclusion

Steganography provides a way to hide information within ordinary digital media. Data hiding provides secrecy, watermarking provides ownership and authenticity, and survivability ensures that the embedded information remains effective after common modifications. Together, they support secure and reliable digital communication.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation